

The T0 Energy Field Model: Mathematical Formulation

Contents

.1	Units Convention	1
	T0 natural units (Heaviside-Lorentz)	1
	Dimensions in natural units	1
.2	Time-Energy Duality	2
	Fundamental relation	2
	Intrinsic time field	2
.3	Universal Field Equation	2
	Wave equation	2
	With sources	2
.4	Lagrangian Density	2
	Universal Lagrangian density	2
	Euler-Lagrange equation	2
.5	Characteristic Energy	2
	Definition	2
	Numerical values	3
	Usage	3
.6	The Parameter ξ	3
	Definition	3
	Geometric components	3
.7	Fine Structure Constant	3
	In T0 units	3
	Reconstruction of SI value	3
	Numerical calculation	4
	Dimension check	4
.8	Gravitational Constant	4
	T0 formula	4
	Fundamental relation	4
.9	Characteristic Lengths	4
	T0 characteristic length	4
	Derivation	5
	Time scale	5
.10	Scale Hierarchy	5
	Planck length as reference	5
	Scale ratio	5
.11	Particles as Field Excitations	5
	Classification by energy	5
	Antiparticles	5
.12	Lepton Masses	5
.13	Anomalous Magnetic Moments	6
.14	Three Field Geometries	6

	Type 1: Localized spherical	6
	Type 2: Localized non-spherical	6
	Type 3: Extended homogeneous	6
.15	Mathematical Identities	6
	Energy field normalization	6
	Duality consistency	6
.16	Dimensional Analysis Verifications	7
	Field equation	7
	Characteristic length	7
	Lagrangian density	7
.17	Formula Reference	7
	Fundamental equations	7
	Derived constants	7
	Characteristic scales	7
	Prediction formulas	7
.18	Numerical Values	8
	Fundamental constants (in natural units)	8
	T0 parameters	8
	Lepton energies	8
	Energy ratios	8
.19	Calculation Examples	8
	Fine structure constant	8
	Characteristic length (electron)	9
0.1	Symbol Reference	9
0.2	Unit Conversions	10
	Natural $\rightarrow$ SI	10
	Standard natural units	10
0.3	Relations to Other Documents	10

Abstract

The T0 model describes physical phenomena through a universal energy field $E_{\text{field}}(x, t)$ with the parameter $\xi = \frac{4}{3} \times 10^{-4}$. The field equation is $\square E_{\text{field}} = 0$, the Lagrangian density $\mathcal{L} = \xi(\partial E)^2$. The model uses standard natural units with $\hbar = c = 1$.

Fundamental quantities:

- Characteristic energy: $E_0 = \sqrt{m_e \cdot m_\mu} = 7.348 \text{ MeV (bare)}$; fractally corrected $E_0/\sqrt{K_{\text{frak}}} = 7.398 \text{ MeV}$ (anchor for α)
- Fine structure constant: $\alpha = \xi(E_0/1 \text{ MeV})^2 \approx 1/137$
- Gravitational constant: $G = \xi^2/(4m_e) \times \text{factors}$

Predictions: Lepton masses with 2% accuracy, fine structure constant $\alpha \approx 1/137$ (from the masses alone $1/137.059, +0.017\%$).

Detailed derivations: See Document 011 (fine structure), 012 (gravitation), 018 (g-2 geometric), 019 (Lagrangian).

.1 Units Convention

T0 natural units (Heaviside-Lorentz)

T0 uses Heaviside-Lorentz natural units:

$$\hbar = c = 4\pi\epsilon_0 = 1 \quad (1)$$

In this system:

- Fine structure constant: $\alpha = e^2 = 1$ (by convention in T0)
- Energy = Mass: $E = m$
- Length = Time = Energy⁻¹: $[L] = [T] = [E^{-1}]$

Note: In SI units, $\alpha \approx 1/137$. The choice $\alpha = 1$ in T0 is a unit convention that simplifies formulas.

Dimensions in natural units

$$[E] = E \quad (2)$$

$$[m] = E \quad (3)$$

$$[t] = E^{-1} \quad (4)$$

$$[L] = E^{-1} \quad (5)$$

$$[G] = E^{-2} \quad (6)$$

$$[\partial_\mu] = E \quad (7)$$

.2 Time-Energy Duality

Fundamental relation

$$T_{\text{field}}(x, t) \cdot E_{\text{field}}(x, t) = 1 \quad (8)$$

with $[T_{\text{field}}] = E^{-1}$ and $[E_{\text{field}}] = E$.

Intrinsic time field

$$T_{\text{field}}(x, t) = \frac{1}{E_{\text{field}}(x, t)} \quad (9)$$

.3 Universal Field Equation

Wave equation

$$\square E_{\text{field}} = 0 \quad (10)$$

with d'Alembert operator:

$$\square = \nabla^2 - \frac{\partial^2}{\partial t^2} \quad (11)$$

With sources

$$\nabla^2 E_{\text{field}} = 4\pi G \rho \cdot E_{\text{field}} \quad (12)$$

Dimension check: $[E^3] = [E^{-2}][E^4][E] = [E^3] \checkmark$

.4 Lagrangian Density**Universal Lagrangian density**

$$\mathcal{L} = \xi \cdot (\partial_\mu E_{\text{field}})(\partial^\mu E_{\text{field}}) \quad (13)$$

with $\xi = \frac{4}{3} \times 10^{-4}$.

Euler-Lagrange equation

$$\frac{\partial \mathcal{L}}{\partial E} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu E)} = 0 \quad (14)$$

yields:

$$\square E_{\text{field}} = 0 \quad (15)$$

.5 Characteristic Energy**Definition**

The characteristic energy E_0 is the geometric mean of electron and muon mass (derivation in Document 011):

$$E_0 = \sqrt{m_e \cdot m_\mu} \quad (16)$$

Numerical values

From experimental masses:

$$E_0 = \sqrt{0.511 \times 105.66} \quad (17)$$

$$= \sqrt{53.99} \quad (18)$$

$$= 7.348 \text{ MeV} \quad (19)$$

Fractally corrected value:

$$E_0^{\text{T0}} = \frac{E_0}{\sqrt{K_{\text{frak}}}} = \frac{7.348}{\sqrt{1 - 100\xi}} = 7.397 \approx 7.398 \text{ MeV} \quad (20)$$

Both values are the same E_0 , bare and fractally corrected: absolute values carry K_{frak} , ratios do not. The square-root factor follows because E_0 enters $\alpha = \xi E_0^2$ quadratically. The corrected value 7.398 MeV serves as the anchor for α .

Usage

E_0 serves as energy scale for:

- Fine structure constant: $\alpha = \xi(E_0/1 \text{ MeV})^2$
- Normalization of electromagnetic effects

.6 The Parameter ξ

Definition

$$\xi = \frac{4}{3} \times 10^{-4} = 1.3333 \times 10^{-4} \quad (21)$$

Dimensionless: $[\xi] = 1$.

Geometric components

$$\xi = G_3 \times S_{\text{ratio}} \quad (22)$$

where:

- $G_3 = \frac{4}{3}$: Geometric factor (sphere-cube ratio)
- $S_{\text{ratio}} = 10^{-4}$: Scale ratio

.7 Fine Structure Constant

In T0 units

In T0 natural units: $\alpha = 1$ (by convention)

Reconstruction of SI value

Important: The SI value can be reconstructed from T0 parameters!

$$\alpha_{\text{SI}} = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2 \quad (23)$$

Numerical calculation

With $\xi = \frac{4}{3} \times 10^{-4}$ and $E_0 = 7.398 \text{ MeV}$:

$$\alpha_{\text{SI}} = 1.3333 \times 10^{-4} \times (7.398)^2 \quad (24)$$

$$= 1.3333 \times 10^{-4} \times 54.73 \quad (25)$$

$$= 7.297 \times 10^{-3} \quad (26)$$

$$= \frac{1}{137.04} \quad (27)$$

Experimental: $\alpha_{\text{exp}} = \frac{1}{137.036}$

Agreement: 0.0005 % (with the anchor $E_0 = 7.398 \text{ MeV}$). From the masses alone, $\alpha = \xi E_0^2 / K_{\text{frak}}$ with $E_0 = 7.348 \text{ MeV}$ gives $\alpha = 1/137.059$ (+0.017 %).

Dimension check

$$[\alpha_{\text{SI}}] = [\xi] \times \left[\frac{E}{E} \right]^2 = 1 \times 1 = 1 \quad \checkmark \quad (28)$$

.8 Gravitational Constant

T0 formula

The gravitational constant is derived from ξ and m_e (derivation in Document 012):

$$G = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \quad (29)$$

where:

- $C_{\text{conv}} = 7.783 \times 10^{-3}$: SI conversion factor
- $K_{\text{frak}} = 0.986$: fractal correction

Fundamental relation

In natural units:

$$\xi = 2\sqrt{G \cdot m_e} \quad (30)$$

Solved for G :

$$G_{\text{nat}} = \frac{\xi^2}{4m_e} \quad (31)$$

Dimension: $[G] = [E^{-2}]$ in natural units; the right-hand side $\xi^2/(4m_e)$, however, has $[E^{-1}]$ – the missing factor $1/E_{\text{char}}$ and the SI conversion reside in C_{conv} (Doc. 012).

.9 Characteristic Lengths

T0 characteristic length

$$r_0 = 2GE \quad (32)$$

Dimension: $[r_0] = [E^{-2}][E] = [E^{-1}] = [L] \checkmark$

Derivation

For spherically symmetric point source $\rho(r) = E_0 \delta^3(\vec{r})$:

Solution of $\nabla^2 E = 4\pi G \rho E$:

$$E(r) = E_0 \left(1 - \frac{r_0}{r}\right) \quad (33)$$

with $r_0 = 2GE_0$.

Time scale

$$t_0 = \frac{r_0}{c} = r_0 = 2GE \quad (34)$$

(since $c = 1$)

.10 Scale Hierarchy

Planck length as reference

$$\ell_P = \sqrt{G} = 1 \quad (\text{in nat. units}) \quad (35)$$

Scale ratio

$$\xi_{\text{ratio}} = \frac{\ell_P}{r_0} = \frac{\sqrt{G}}{2GE} = \frac{1}{2\sqrt{G} \cdot E} \quad (36)$$

For $E \sim 1$ GeV:

$$\frac{r_0}{\ell_P} = 2\sqrt{G}E = \frac{2E}{E_P} \sim 1.6 \times 10^{-19} \quad (\text{sub-Planck}) \quad (37)$$

.11 Particles as Field Excitations

Classification by energy

Particle	Energy [MeV]
Electron	0.511
Muon	105.658
Tau	1776.86

Antiparticles

Negative field excitations: $E_{\text{field}} < 0$

.12 Lepton Masses

The T0 model predicts lepton masses (derivation in Document 003; mass ladder $m_i = r_i \xi^{p_i} \nu$ following Docs. 005 and 006):

Lepton	T0 [MeV]	Exp [MeV]	Δ [%]
Electron	0.505	0.511	-1.18
Muon	104.96	105.66	-0.66
Tau	1783.4	1776.9	+0.37

.13 Anomalous Magnetic Moments

Anomalous magnetic moments are treated in dedicated documents: Doc. 018 (with its revisions) and Doc. 158.

.14 Three Field Geometries

Type 1: Localized spherical

$$E(r) = E_0 \left(1 - \frac{\beta}{r}\right), \quad \beta = r_0 \quad (38)$$

Application: Individual particles (electron, muon, tau)

Type 2: Localized non-spherical

$$E(\vec{r}) = E_0 \left(1 - \frac{\beta_{ij} r_i r_j}{r^3} \right) \quad (39)$$

Application: Bound systems

Type 3: Extended homogeneous

Effective parameter:

$$\xi_{\text{eff}} = \frac{\xi}{2} = \frac{2}{3} \times 10^{-4} \quad (40)$$

Application: Cosmology (see Document 026)

.15 Mathematical Identities**Energy field normalization**

$$E_{\text{field}}(\vec{r}, t) = E_0 \cdot f(\vec{r}, t) \cdot e^{i\phi(\vec{r}, t)} \quad (41)$$

with:

- E_0 : Characteristic energy
- $f(\vec{r}, t)$: Normalized space-time profile of the energy field (not a frequency and not the base winding number $f = 1/\xi$)
- $\phi(\vec{r}, t)$: Phase

Duality consistency

Time-mass (Document 003): $T \cdot m = 1$

Time-energy (this document): $T \cdot E = 1$

In natural units ($c = 1$):

$$E = mc^2 = m \quad \Rightarrow \quad T \cdot m = T \cdot E \quad (42)$$

.16 Dimensional Analysis Verifications**Field equation**

$$[\nabla^2 E] = [L^{-2}][E] = [E^2][E] = [E^3] \quad (43)$$

$$[4\pi G \rho E] = [E^{-2}][E^4][E] = [E^3] \quad \checkmark \quad (44)$$

Characteristic length

$$[r_0] = [2GE] = [E^{-2}][E] = [E^{-1}] = [L] \quad \checkmark \quad (45)$$

Lagrangian density

$$[\mathcal{L}] = [\xi][(\partial E)^2] = [1][E^4] = [E^4] \quad (\text{correct for Lagrangian density}) \quad (46)$$

.17 Formula Reference

Fundamental equations

$$\text{Duality: } T_{\text{field}} \cdot E_{\text{field}} = 1 \quad (47)$$

$$\text{Wave equation: } \square E_{\text{field}} = 0 \quad (48)$$

$$\text{With sources: } \nabla^2 E = 4\pi G \rho E \quad (49)$$

$$\text{Lagrangian density: } \mathcal{L} = \xi(\partial E)^2 \quad (50)$$

Derived constants

$$\text{Characteristic energy: } E_0 = \sqrt{m_e \cdot m_\mu} = 7.348 \text{ MeV (bare; corrected 7.398)} \quad (51)$$

$$\text{Fine structure constant: } \alpha = \xi(E_0/1 \text{ MeV})^2 \approx 1/137 \quad (52)$$

$$\text{Gravitational constant: } G = \frac{\xi^2}{4m_e} \times \text{factors} \quad (53)$$

Characteristic scales

$$\text{T0 length: } r_0 = 2GE \quad (54)$$

$$\text{T0 time: } t_0 = 2GE \quad (55)$$

$$\text{Planck length: } \ell_P = \sqrt{G} = 1 \quad (56)$$

$$\text{Scale ratio: } \xi_{\text{ratio}} = \frac{1}{2\sqrt{GE}} \quad (57)$$

Prediction formulas

$$\text{Parameter: } \xi = \frac{4}{3} \times 10^{-4} \quad (58)$$

$$\text{Effective parameter: } \xi_{\text{eff}} = \frac{\xi}{2} \quad (59)$$

.18 Numerical Values

Fundamental constants (in natural units)

$$\hbar = 1 \quad (60)$$

$$c = 1 \quad (61)$$

$$\alpha = \frac{1}{137.036} \approx 7.297 \times 10^{-3} \quad (62)$$

$$G = 1 \text{ (numerically, dimension } [E^{-2}]) \quad (63)$$

T0 parameters

$$\xi = \frac{4}{3} \times 10^{-4} = 1.3333 \times 10^{-4} \quad (64)$$

$$\xi^2 = 1.7778 \times 10^{-8} \quad (65)$$

$$\frac{\xi}{2\pi} = 2.1221 \times 10^{-5} \quad (66)$$

$$\xi_{\text{eff}} = 6.6667 \times 10^{-5} \quad (67)$$

$$E_0 = 7.348 \text{ MeV (from exp. masses)} \quad (68)$$

$$E_0^{\text{T0}} = 7.398 \text{ MeV (fractally corrected, } E_0/\sqrt{K_{\text{frak}}}) \quad (69)$$

Lepton energies

$$E_e = 0.511 \text{ MeV} \quad (70)$$

$$E_\mu = 105.658 \text{ MeV} \quad (71)$$

$$E_\tau = 1776.86 \text{ MeV} \quad (72)$$

Energy ratios

$$\frac{E_\mu}{E_e} = 206.768 \quad (73)$$

$$\frac{E_\tau}{E_e} = 3477.2 \quad (74)$$

$$\frac{E_\tau}{E_\mu} = 16.817 \quad (75)$$

.19 Calculation Examples

Fine structure constant

Given:

- $\xi = 1.3333 \times 10^{-4}$
- $E_0 = 7.398 \text{ MeV}$

Calculation:

$$\left(\frac{E_0}{1 \text{ MeV}} \right)^2 = (7.398)^2 = 54.73 \quad (76)$$

$$\alpha = 1.3333 \times 10^{-4} \times 54.73 \quad (77)$$

$$= 7.297 \times 10^{-3} \quad (78)$$

$$= \frac{1}{137.04} \quad (79)$$

Experimental: $\alpha_{\text{exp}} = \frac{1}{137.036}$

Deviation: 0.0005 %

Characteristic length (electron)

Given:

- $E_e = 0.511 \text{ MeV} = 0.511 \times 1.6 \times 10^{-13} \text{ J} = 8.2 \times 10^{-14} \text{ J}$
- $G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
- $c = 3 \times 10^8 \text{ m/s}$

Conversion to SI units:

$$r_0 = \frac{2GE}{c^4} \approx 1.35 \times 10^{-57} \text{ m} \quad (80)$$

Planck comparison:

$$\frac{r_0}{\ell_P} = \frac{1.35 \times 10^{-57}}{1.6 \times 10^{-35}} \approx 8.4 \times 10^{-23} \quad (81)$$

0.1 Symbol Reference

Symbol	Meaning	Dimension
ξ	Fundamental parameter	1
E_0	Characteristic energy	E
E_{field}	Universal energy field	E
T_{field}	Intrinsic time field	E^{-1}
r_0	T0 characteristic length	$L = E^{-1}$
t_0	T0 characteristic time	$T = E^{-1}$
ℓ_P	Planck length	$L = E^{-1}$
G	Gravitational constant	E^{-2}
α	Fine structure constant	1
E_e, E_μ, E_τ	Lepton energies	E
m_e, m_μ, m_τ	Lepton masses (= E in nat. units)	E
$\mathcal{L}$	Lagrangian density	E^4
$\square$	d'Alembert operator	E^2
ξ_{eff}	Effective parameter ($\xi/2$)	1

0.2 Unit Conversions

Natural $\rightarrow$ SI

$$1 \text{ (Energy)} = 1 \text{ GeV} = 1.6 \times 10^{-10} \text{ J} \quad (82)$$

$$1 \text{ (Length)} = \frac{\hbar c}{1 \text{ GeV}} = 0.197 \text{ fm} \quad (83)$$

$$1 \text{ (Time)} = \frac{\hbar}{1 \text{ GeV}} = 6.58 \times 10^{-25} \text{ s} \quad (84)$$

Standard natural units

In standard convention ($\hbar = c = 1$):

- $\alpha = \frac{e^2}{4\pi\epsilon_0} \approx \frac{1}{137}$ (dimensionless)
- All quantities in powers of energy
- Physical predictions identical to other conventions

0.3 Relations to Other Documents

- **Document 003:** Time-mass duality, foundations, origin of ξ
- **Document 011:** Fine structure constant (geometric derivation)
- **Document 012:** Gravitational constant (systematic derivation)
- **Document 018:** Geometric g-2 formulation (fractal geometry)
- **Document 019:** Lagrangian formulation (quantum field theory)
- **Document 026:** Cosmology ($\xi_{\text{eff}} = \xi/2$)
All formulations are based on $\xi = \frac{4}{3} \times 10^{-4}$.